

Report R18 — two conservation laws, or frozen domain walls? Rules 108/201 and 156/198

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

Hilbert-space fragmentation is often explained by the interaction of two conservation laws — charge and dipole moment, or domain walls and magnetisation (Khemani, Hermele and Nandkishore, arXiv:1912.04300). We test that account against the four rules R9 places off both axes of its map. The verdict is that the fragmentation is **fully constructive from frozen domain walls**, and that the conservation laws, which do exist and are exactly of the promised type, are *consequences* of the wall grammar rather than an independent ingredient. In detail: (i) there is no charge at all in the computational basis — magnetisation is not conserved, so the charge/dipole ladder never starts; (ii) there is exactly one conserved domain-wall charge per rule, and the two families differ in which one — W_{108}/W_{201} conserve the *staggered* wall number $S = \sum_i (-1)^i b_i$ while W_{156}/W_{198} conserve the *total* wall number $D = \sum_i b_i$, with the other broken in each case; (iii) taking both independent charges together their joint level sets number 36 at $N = 14$ against 3329 Krylov sectors, so they are short of the required cardinality by an exponentially growing factor and cannot be the explanation; (iv) each rule has exactly *one* minimal wall word — 00, 11, 01, 10 — and the *set of its occurrences* is a complete invariant, verified exhaustively to $N = 14$; (v) the wall partition reproduces not merely the sector count but the entire multiset of sector sizes, so both of R9's exponents fall out of it, $\rho^2 = 1.754878$ from an exact order-3 integer recurrence and φ from Fibonacci; and (vi) the conserved charge is a *function* of the wall set — for the chiral pair simply $D = 2|W|$ — so it carries strictly less information than the walls do.

Contents

1	The four rules and the question	1
2	The conservation laws exist, and they are domain-wall numbers	2
3	But they are far too coarse to explain the fragmentation	2
4	The wall grammar is enough	3
4.1	One word per rule	3
4.2	The occurrence set is a complete invariant	3
4.3	Both exponents fall out of it	4
4.4	Two details that are not decoration	4
5	The charge is a shadow of the grammar	4
6	Answering the question as asked	5
7	Reproduction	5

1 The four rules and the question

All four rules are unitary, so by R17 their terminal SCCs are their Krylov sectors, and all four carry a single Hadamard whose gate fires on one neighbour pattern:

rule	tuple	fires when $(x_{i-1}, x_{i+1}) =$	family
W_{201}	VIII	$(0, 0)$	the PXP constraint
W_{108}	IIIV	$(1, 1)$	its spin-flip image
W_{156}	IIVI	$(1, 0)$	chiral
W_{198}	IVII	$(0, 1)$	its reflection

R9 reports sector-count bases $\rho^2 = 1.754878$ for the first pair and φ for the second, with ρ the plastic number. The question is whether that fragmentation is the shadow of two conserved densities, as in the dipole-conserving models, or whether it is generated constructively by frozen walls.

A note on where the tools came from. The wall grammar and the charge null-space detector are Tier-2 modules (`qca/walls.py`, `qca/charges.py`); they were run read-only for this report. The wall detector in `scaling/wall_charges.py` is an *independent* Tier-1 reimplementa- tion, and a regression test pins its output against the four words Tier-2 reports, so the central claim rests on two separate implementations agreeing. The two engines were also checked against each other on the sector counts and largest-sector sizes of all four rules across $N = 6 \dots 14$: they agree term for term.

2 The conservation laws exist, and they are domain-wall num- bers

Write $b_i = x_i \oplus x_{i+1}$ for the bond (domain-wall) variables, and

$$D = \sum_i b_i, \quad S = \sum_i (-1)^i b_i.$$

Proposition 1. *For W_{108} and W_{201} , S is conserved on every sector and D is not. For W_{156} and W_{198} , D is conserved and S is not. This holds at both *obc0* and *pbc* for every N tested.*

Two further facts frame these. First, Tier-2’s null-space detector finds **no charge at all at range 1 in the computational basis** for any of the four rules: single-site magnetisation is not conserved, which is immediate once one notices that the gate flips a site outright. So the “charge plus dipole” construction has nothing to build on here; the only conserved densities live in the bond basis. Second, the rank of the certified charge values on the full 2^N space is 3 including the constant, i.e. there are exactly **two** independent local conserved quantities per rule — the right number for the two-conservation-law story, which is precisely why the story deserves testing rather than dismissal.

3 But they are far too coarse to explain the fragmentation

rule	conserved	N	levels (wall charge)	levels (both)	Krylov sectors	ratio
W_{108}	S, —W—	6	4	10	37	3.7
W_{108}	S, —W—	8	5	15	114	7.6
W_{108}	S, —W—	10	6	21	351	16.7
W_{108}	S, —W—	12	7	28	1081	38.6
W_{108}	S, —W—	14	8	36	3329	92.5
W_{201}	S, —W—	6	4	9	21	2.3
W_{201}	S, —W—	8	5	14	65	4.6
W_{201}	S, —W—	10	6	20	200	10.0
W_{201}	S, —W—	12	7	27	616	22.8
W_{201}	S, —W—	14	8	35	1897	54.2
W_{156}	D, —W—	6	4	4	21	5.2
W_{156}	D, —W—	8	5	5	55	11.0
W_{156}	D, —W—	10	6	6	144	24.0
W_{156}	D, —W—	12	7	7	377	53.9
W_{156}	D, —W—	14	8	8	987	123.4
W_{198}	D, —W—	6	4	4	21	5.2
W_{198}	D, —W—	8	5	5	55	11.0
W_{198}	D, —W—	10	6	6	144	24.0
W_{198}	D, —W—	12	7	7	377	53.9
W_{198}	D, —W—	14	8	8	987	123.4

Testing the hypothesis fairly means using *both* independent charges, not just the domain-wall one: the second is the wall count $|W|$, a range-2 charge in the computational basis. For W_{108} their joint level sets number 10, 15, 21, 28, 36 over $N = 6, 8, 10, 12, 14$ — visibly quadratic — while the sectors number 37, 114, 351, 1081, 3329. The ratio is therefore not merely large but *exponentially growing*: 3.7, 7.6, 16.7, 38.6, 92.5.

For W_{156}/W_{198} the two charges coincide, since $D = 2|W|$ identically (§5), so the table’s level count for them is a *lower* bound on what the symmetry resolves and the second independent charge Tier-2’s detector reports is not identified here. It does not matter for the conclusion: running Tier-2’s full certified charge set on the chiral pair gives 12, 17, 23 level sets at $N = 8, 10, 12$ against 48, 124, 323 sectors, i.e. ratios 4.0, 7.3, 14.0 — diverging just the same.

This is the standard diagnostic of fragmentation — exponentially many Krylov sectors inside polynomially many symmetry sectors — and it is exactly what rules out the two-conservation-law account as an *explanation*. Two charges of this kind can label $O(N^2)$ things. They cannot label ρ^{2N} things. Whatever is doing the shattering is not them.

4 The wall grammar is enough

4.1 One word per rule

Detection is the Tier-2 criterion: a word w is a wall if, placed on the chain at either sublattice offset, its sites are invariant under one full cycle for *every* exterior configuration. Since one cycle has light cone 2, two context sites on each side exhaust the exterior dependence and the sixteen configurations of those four sites are the whole test. Each of the four rules has exactly one minimal wall word:

$$W_{108} : 00, \quad W_{201} : 11, \quad W_{156} : 01, \quad W_{198} : 10.$$

For W_{201} this is transparent: a site flips only when both its neighbours are 0, so two adjacent 1s can never move, and neither can anything they flank.

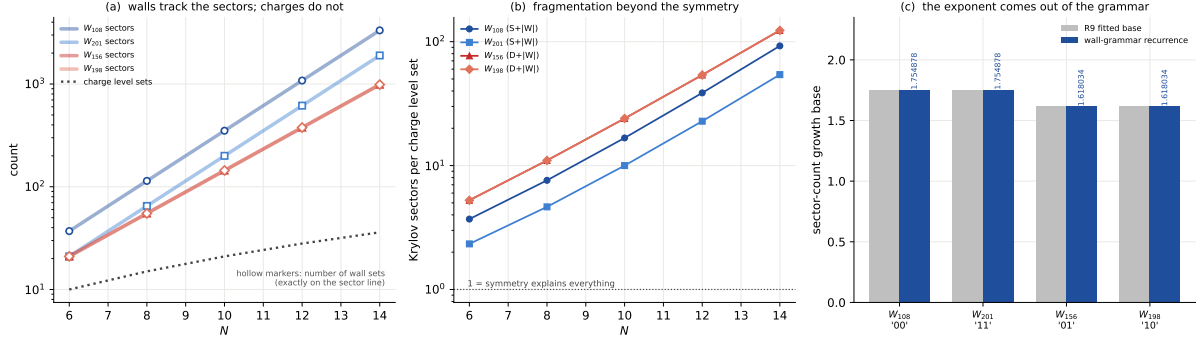


Figure 1: **Walls explain the sectors; the charges cannot.** (a) Sector count (thick line) against the number of distinct wall-occurrence sets (hollow markers) — they coincide exactly at every N — and against the number of joint level sets of the conserved charges (dotted), which stays in single figures. (b) Krylov sectors per symmetry sector: exponentially growing, so the symmetry cannot be the explanation; the dotted line at 1 is where a purely symmetry-resolved model would sit. (c) The dominant root of the exact integer recurrence satisfied by the number of wall sets, against the growth base R9 fitted for the sector count — agreement to six digits, with the exponent derived rather than fitted.

4.2 The occurrence set is a complete invariant

Proposition 2. *Two states lie in the same Krylov sector if and only if the rule’s wall word occurs at the same set of positions. Verified exhaustively for all four rules at $obc0$ for $N \leq 14$, and at pbc for W_{108} and W_{201} .*

Both halves matter and are reported separately in `wall_completeness`: the wall set is *constant* on each sector (it is conserved) and *injective* across sectors (it separates them). What makes this a constructive account rather than a restatement is that the label is computed from a single state, with no reference to the sector it lies in. By contrast the “frozen-site signature” — the positions and values of the sites that happen to be constant across a sector — is also a complete invariant for all four rules, but it is nearly vacuous, because it is defined *from* the sector. We record it only to note that it is not the content of this section.

rule	tuple	wall	conserved	N	sectors	wall sets	complete	sizes	rec. order	root	R9 base
W_{108}	IIIV	00	S	14	3329	3329	yes	yes	3	1.754878	1.754878
W_{201}	VIII	11	S	14	1897	1897	yes	yes	3	1.754878	1.754878
W_{156}	IIVI	01	D	14	987	987	yes	yes	2	1.618034	1.618034
W_{198}	IVII	10	D	14	987	987	yes	yes	2	1.618034	1.618034

4.3 Both exponents fall out of it

The number of realisable wall sets satisfies an exact homogeneous linear recurrence with integer coefficients, found by exact rational elimination rather than by fitting:

$$\begin{aligned}
 W_{108}, W_{201} : \quad a(N) &= 2a(N-1) - a(N-2) + a(N-3), & \text{dominant root } 1.754878 = \rho^2, \\
 W_{156}, W_{198} : \quad a(N) &= a(N-1) + a(N-2), & \text{dominant root } 1.618034 = \varphi.
 \end{aligned}$$

These reproduce R9’s fitted sector-count bases to six digits (Fig. 1c). Moreover the wall partition gives the *whole* size distribution, not just the count: the multiset of preimage sizes of the map $x \mapsto W(x)$ equals the multiset of sector sizes at every N tested, so $D_{\max}$ comes out too — Fibonacci for W_{108}/W_{201} , and R9’s room-packing series 6, 9, 12, 16, 20, 27, 36, 48, 64, ... for W_{156}/W_{198} .

4.4 Two details that are not decoration

The boundary is wall material. At `obc0` the chain is padded with a frozen 0 outside each end, and the wall word must be searched on the padded string. This is not bookkeeping: W_{108} 's word is 00, the padding supplies 0s, and searching the bare chain finds only 616 of the 1081 sectors at $N = 12$. An earlier pass of this analysis concluded from the bare search that W_{156}/W_{198} needed the charge *in addition* to the walls; that was an artefact of the missing padding and is not the case.

One defect, on the ring. For W_{156}/W_{198} at `pbcc` the wall set is complete except for a single collision at every N : the two uniform states 0^N and 1^N are both frozen singletons and neither contains 01, so they share the empty wall set and $n_{\text{sectors}} = n_{\text{wall sets}} + 1$ exactly (48 vs 47, 124 vs 123, 323 vs 322, 844 vs 843). W_{108}/W_{201} have no such defect, because for them 0^N and 1^N are not both frozen.

5 The charge is a shadow of the grammar

The decisive step is not that the walls suffice but that the charges are *downstream* of them.

Proposition 3. *For all four rules, at both boundary conditions and every N tested, the conserved charge is a function of the wall set. For W_{156} and W_{198} it is explicitly*

$$D = 2|W|,$$

$|W|$ being the number of occurrences of the wall word.

The relation for the chiral pair is transparent once stated: on a ring the 01 and 10 occurrences are equinumerous, and D counts both, so the total domain-wall number is exactly twice the number of walls. For W_{108}/W_{201} the staggered charge S is likewise determined by the wall set but is not affine in $|W|$ — it depends on the parities of the wall positions, which is what “staggered” means.

So the implication runs one way only: the wall configuration determines the charge, the charge does not determine the wall configuration, and the sectors are the level sets of the former. The conservation law is a coarse-grained readout of the wall data, not a second ingredient interacting with it.

6 Answering the question as asked

1. **Is it explained by the interaction of two conservation laws?** No. There are exactly two independent local conserved quantities, which is the right number, and one of them is genuinely a domain-wall charge, which is the right kind. But their joint level sets number $O(N^2)$ — 36 of them at $N = 14$ — against 3329 Krylov sectors, and the ratio grows from 3.7 to 92.5 over $N = 6 \dots 14$. They do not have the cardinality to be the explanation.
2. **Is it fully constructive from the frozen domain walls?** Yes. One minimal wall word per rule; the set of its occurrences is a complete invariant, computed locally from a single state; the induced partition reproduces the sector count, the full size distribution and both of R9's exponents, with the exponents coming from exact integer recurrences rather than fits. The single exception is the two uniform states on the ring for the chiral pair.
3. **And the relation between the two accounts?** The charges are functions of the wall set — $D = 2|W|$ for the chiral pair — so the constructive account strictly subsumes

the symmetry account. This is the opposite of the dipole-conserving picture, where the charges are given and the shattering is an emergent extra; here the walls are given and the charges are the emergent extra.

4. **An incidental contrast worth keeping.** Which wall-number charge survives is exactly what separates the two families: the PXP-like pair conserves the staggered wall number and the chiral pair the total one, and that difference tracks the difference in their exponents (ρ^2 against φ).

7 Reproduction

```
python -m qca_fragmentation.scaling.wall_charges --bc obc0 --rebuild
python -m pytest qca_fragmentation/tests/test_wall_charges.py -q
```

Everything here comes from `analytics/wall_charges_obc0.json`. The Tier-2 detectors used for the cross-checks in §2 — the null-space charge detector, its gate-local certificate, and the wall grammar — live in the Tier-2 repository and were not modified; only read. If this analysis is to be adopted there, `scaling/wall_charges.py` depends on Tier-1 only for `core.cycle.succ`, `core.rules` and the union-find in `graph.wcc.make_succ`, all of which Tier-2 has equivalents for.